RBF-ESO Backstepping Control for Multi-UAV Cooperative Payload Transportation

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Abstract

Cooperative payload transportation with multiple unmanned aerial vehicles (UAVs) is challenging because the payload dynamics are strongly coupled with cable constraints, nonlinear rotational motion, allocation errors, and external disturbances. This paper presents a payload-level RBF-ESO backstepping framework in which the RBF output is placed inside the ESO prediction model, so that the observer estimates the residual that remains after learning instead of a separately defined disturbance. One common parameter set is frozen for every configuration, and the evidence is organised as a backstepping-family ablation together with a mass/inertia mismatch sweep, an unmodeled quadratic drag, and a band-limited non-periodic disturbance experiment. Across these conditions the ESO-only configuration accounts for most of the error reduction (about thirteen times relative to plain backstepping under the periodic disturbance and about eight times under the random one), whereas the additional RBF term contributes a consistent but modest 6–10%. The ordering of the four configurations is preserved under every disturbance class tested. The paper therefore bounds the role that can be attributed to each module rather than claiming that the network learns the unmodeled dynamics.

Keywords
multi-UAV cooperative transportation; payload transportation; backstepping control; RBF neural network; extended state observer; non-periodic disturbance; ablation

Introduction

UAV technology has developed rapidly in recent years and is now widely used in aerial imaging, agriculture, industry, and material delivery. Recently, increasing attention has been directed toward multi-UAV systems. Compared with single-UAV platforms, multi-UAV cooperative systems provide higher payload capacity, greater flexibility, and improved fault tolerance. Among these applications, multi-UAV cooperative payload transportation has attracted significant interest due to its potential for transporting heavy and large-sized materials (Fink et al. 2011; Lee et al. 2013c; Estevez et al. 2024).

However, cooperative payload transportation remains a challenging problem in aerial robotics. The presence of flexible cables, coupled motion constraints, and strong interactions among multiple UAVs and the payload leads to a high-dimensional, nonlinear, and tightly coupled dynamical system. In practice, these couplings introduce significant difficulties in both controller design and stability analysis, especially when precise trajectory tracking and attitude regulation are required simultaneously (Goodarzi et al. 2015; Sreenath et al. 2013; Lee 2018). Although a complete geometric dynamic model has been established in prior work (Lee et al. 2013a; Lee 2014), achieving robust and high-accuracy control under external disturbances, parameter uncertainties, and unmodeled dynamics is still nontrivial. Therefore, developing control strategies that combine accurate modeling with strong disturbance-rejection capability remains an important research direction for cooperative multi-UAV payload transportation.

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To address these challenges, a variety of control strategies have been developed in the literature. Nonlinear methods, such as backstepping and sliding-mode control, have been widely adopted for UAV systems because of their ability to handle strong nonlinearities (Bouabdallah and Siegwart 2005; Madani and Benallegue 2008; Thanh et al. 2022). Nevertheless, these approaches often depend on relatively accurate system models, and their performance can degrade when significant parameter uncertainties and unmodeled dynamics are present. To mitigate these limitations, adaptive control schemes and neural-network-based methods have been introduced to approximate unknown dynamics online (Sanner and Slotine 1992; Park and Sandberg 1991; Shen and Xu 2021). In addition, ESO and ADRC methods provide a practical observer-based route for estimating lumped disturbances and improving robustness (Han 2009; Gao 2003; Madoński and Herman 2015; Wang et al. 2022). These developments motivate the use of learning-based compensation terms together with robust feedback to improve tracking performance under modeling uncertainty.

In this paper, we develop an RBF-ESO backstepping controller for multi-UAV cooperative payload transportation. Backstepping is adopted as the nominal nonlinear tracking controller because it provides a recursive error-stabilization structure for the payload translation and attitude channels. A bank of small RBF networks supplies an additional online compensation term (Broomhead and Lowe 1988; Park and Sandberg 1991; Shen and Xu 2021), and an ESO estimates the lumped residual that remains *after* that compensation (Han 2009; Madoński and Herman 2015; Gu et al. 2022). The structural point is that the RBF output enters the ESO prediction model: the observer therefore tracks a residual that changes while the network learns, instead of estimating a disturbance that is defined independently of the learning module.

The contribution of this paper is a controlled ablation framework together with the residual-consistent architecture it evaluates. The framework freezes one common parameter set for all variants, reports the marginal value of each module, and states explicitly which conclusions the numerical evidence does and does not support. The evidence covers a periodic multi-harmonic disturbance, a band-limited non-periodic disturbance over five fixed seeds, mass and inertia mismatches from 0.5 to 1.5 times the nominal values, and an unmodeled quadratic drag

reaching about 74% of the payload weight. The resulting statement is deliberately narrow: the observer carries most of the compensation, the learning module adds a consistent but modest margin, and the ordering of the four configurations is preserved across the disturbance classes tested.

System Modeling

The dynamic model used in this paper is based on the geometric formulation proposed by Lee et al. for multiple quadrotor UAVs transporting a cable-suspended rigid body (Lee et al. 2013a,c). In that framework, a cooperative aerial transportation system consists of N quadrotors carrying a rigid payload through massless, taut, inextensible cables. The payload dynamics and individual quadrotor dynamics are derived separately and are coupled through kinematic constraints.

Coordinate Frames and State Representation

We define three primary coordinate frames:

Inertial Frame (I) A fixed Earth frame defined by basis vectors $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, with $\mathbf{e}_3 = [0, 0, 1]^T$ pointing **upward**; gravity therefore enters the translational dynamics as $-g\mathbf{e}_3$, which is the convention used consistently in the model, the controller, and the implementation.

Payload Body Frame (L) Attached to the payload center of mass (CoM), with orientation $\mathbf{R}_L \in SO(3)$ relative to I.

Quadrotor Body Frame (B_i) Attached to the CoM of the i-th quadrotor ($i = 1 \dots N$), with orientation $\mathbf{R}_i \in SO(3)$ relative to I.

For both the payload and each quadrotor, the dynamic variables are position $\mathbf{p}$, linear velocity $\mathbf{v}$, attitude matrix $\mathbf{R}$, and angular velocity $\mathbf{\Omega}$. Here, $\mathbf{p}$ and $\mathbf{v}$ are expressed in the inertial frame, $\mathbf{R} \in SO(3)$ denotes attitude, and $\mathbf{\Omega}$ is expressed in the corresponding body frame. Subscripts denote object indices: L for the payload, i for the i-th quadrotor (or cable), and ext for external disturbances.

Kinematic Constraints

The payload is connected to the i-th quadrotor by a taut, inextensible cable of length l_i. Let $\boldsymbol{\rho}_i$ be the vector of the connection point on the load in the L frame, then the geometric relationship follows the equation below.

$$\mathbf{p}_i = \mathbf{p}_L + \mathbf{R}_L \boldsymbol{\rho}_i - l_i \mathbf{q}_i \tag{1}$$

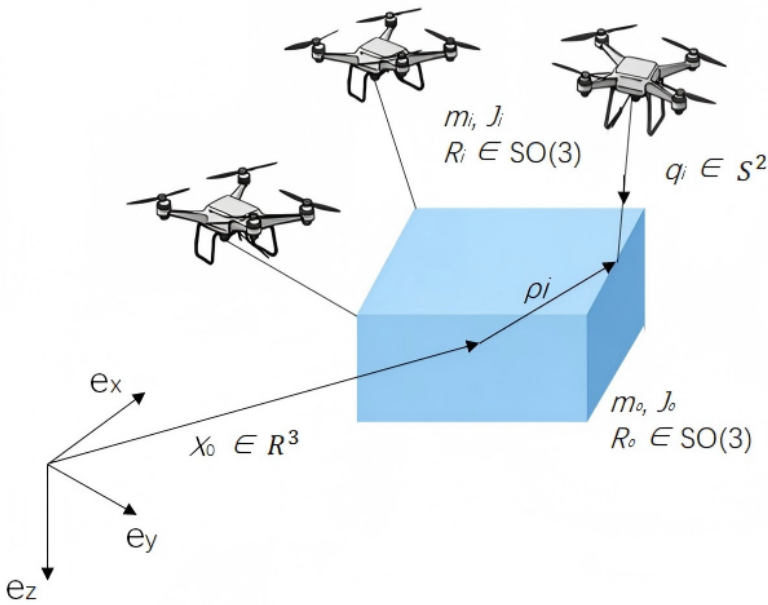


Figure 1. The UAVs and payload geometric structure.

where $\mathbf{q}_i \in S^2$ is the unit vector representing the cable direction, pointing from the quadrotor to the payload.

Differentiating with respect to time yields the velocity relationship

$$\mathbf{v}_i = \mathbf{v}_L + \mathbf{R}_L(\boldsymbol{\Omega}_L \times \boldsymbol{\rho}_i) - l_i\boldsymbol{\omega}_i \times \mathbf{q}_i, \tag{2}$$

where the cable angular velocity satisfies $\dot{\mathbf{q}}_i = \boldsymbol{\omega}_i \times \mathbf{q}_i$.

Payload Dynamics

We consider a 6-DOF rigid body with mass m_L and inertia tensor $\mathbf{J}_L$, and derive the equations of motion using the Newton–Euler method.

Translational Dynamics The total force acting on the payload is composed of gravity, the tension forces $\mathbf{T}_i$ from all N cables, and an external disturbance force $\mathbf{f}_{ext}$:

$$m_L\dot{\mathbf{v}}_L = -m_L g\mathbf{e}_3 + \sum_{i=1}^{N}\mathbf{T}_i + \mathbf{f}_{ext}, \tag{3}$$

where $\mathbf{T}_i = -\mu_i \mathbf{q}_i$ is the cable force and $\mu_i > 0$ is the corresponding tension magnitude, so that the signed tension indicator used later is $T_i = -\mathbf{q}_i^T \boldsymbol{\mu}_i$ with $\boldsymbol{\mu}_i$ the force actually applied by the i-th cable (the signed indicator is defined in "Control Allocation and Cable Force Generation").

Rotational Dynamics The rotational motion is driven by the moments generated by the cable forces applied at lever arms $\boldsymbol{\rho}_i$:

$$\mathbf{J}_L \dot{\boldsymbol{\Omega}}_L = -\boldsymbol{\Omega}_L \times (\mathbf{J}_L \boldsymbol{\Omega}_L) + \sum_{i=1}^{N} \boldsymbol{\rho}_i \times (\mathbf{R}_L^T \mathbf{T}_i) + \mathbf{J}_L \mathbf{d}_R, \tag{4}$$

where $\mathbf{d}_R \in \mathbb{R}^3$ is an external *angular acceleration* (rad/s^2), which is how the disturbance is injected in the implementation; an equivalent disturbance moment is $\mathbf{J}_L \mathbf{d}_R$.

Quadrotor Dynamics

Each quadrotor is treated as an independent rigid body subject to its own thrust, gravity, and the reaction force from the cable tension ($-\mathbf{T}_i$).

Translational Dynamics

$$m_i \dot{\mathbf{v}}_i = -m_i g \mathbf{e}_3 - f_i \mathbf{R}_i \mathbf{e}_3 - \mathbf{T}_i, \tag{5}$$

where f_i is the total scalar thrust generated by the i-th quadrotor rotors, and $-\mathbf{R}_i \mathbf{e}_3$ represents the thrust direction in the inertial frame.

Rotational Dynamics

$$\mathbf{J}_i \dot{\boldsymbol{\Omega}}_i = -\boldsymbol{\Omega}_i \times (\mathbf{J}_i \boldsymbol{\Omega}_i) + \boldsymbol{\tau}_i, \tag{6}$$

where $\boldsymbol{\tau}_i = [\tau_\phi, \tau_\theta, \tau_\psi]^T$ are the control torques in the body frame.

Hierarchical Control Design

To address uncertainties in payload dynamics and external disturbances, we propose an adaptive robust control scheme that integrates backstepping control, radial basis function (RBF) neural-network approximation, and extended state observer (ESO) disturbance estimation while following the geometric payload-transport control framework in (Lee et al. 2013a; Lee 2018). The backstepping component generates the nominal payload force and moment commands, the RBF bank compensates part of the unmodeled nonlinear dynamics (Sanner and Slotine 1992; Park and

Sandberg 1991), and the ESO estimates the residual that the network does not capture (Han 2009; Madoński and Herman 2015).

The overall signal flow is shown in Fig. 2. The payload tracking errors are processed by the nominal backstepping structure, while the RBF and ESO modules provide online compensation. The resulting payload-level force and moment commands are distributed through the cable-force allocation layer and converted into executable quadrotor thrust and attitude commands. Note that the observer is driven by the wrench that the allocation layer actually applies, not by the commanded wrench.

State and Error Definitions

The payload translational state is described by the position $\mathbf{p}_L \in \mathbb{R}^3$ and linear velocity $\mathbf{v}_L \in \mathbb{R}^3$, with desired trajectories $\mathbf{p}_{L,d}$ and $\mathbf{v}_{L,d}$. The payload attitude can be equivalently represented by the rotation matrix $\mathbf{R}_L \in SO(3)$ for geometric modeling, while the implemented controller uses ZYX Euler angles. Let

$$\boldsymbol{\Theta}_L = \begin{bmatrix} \phi_L \\ \theta_L \\ \psi_L \end{bmatrix}, \qquad \boldsymbol{\Theta}_{L,d} = \begin{bmatrix} \phi_{L,d} \\ \theta_{L,d} \\ \psi_{L,d} \end{bmatrix}, \tag{7}$$

denote the current and desired Euler-angle vectors, and let $\dot{\boldsymbol{\Theta}}_L = \mathbf{T}(\boldsymbol{\Theta}_L)\boldsymbol{\Omega}_L$ be the Euler-angle rate, with $\mathbf{T}(\cdot)$ the ZYX kinematic matrix and $\dot{\mathbf{T}}$ its time derivative. The tracking errors used in the controller are

$$\begin{aligned} \mathbf{e}_{p_L} &= \mathbf{p}_L - \mathbf{p}_{L,d}, & \mathbf{e}_{v_L} &= \mathbf{v}_L - \dot{\mathbf{p}}_{L,d}, \\ \mathbf{e}_{\Theta_L} &= \mathrm{wrap}(\boldsymbol{\Theta}_L - \boldsymbol{\Theta}_{L,d}), & \mathbf{e}_{\dot{\Theta}_L} &= \dot{\boldsymbol{\Theta}}_L - \dot{\boldsymbol{\Theta}}_{L,d}, \end{aligned} \tag{8}$$

where $\mathrm{wrap}(\cdot)$ maps each angle difference to $(-\pi, \pi]$ so that a crossing of the $\pm\pi$ boundary does not produce a spurious error.

The RBF banks are *channel-wise*: channel i receives its own two-dimensional feature vector

$$\boldsymbol{\xi}_i = \begin{bmatrix} e_i \\ \dot{e}_i \end{bmatrix} \in \mathbb{R}^2, \qquad i = 1, \ldots, 6, \tag{9}$$

where e_i, $i = 1, 2, 3$, are the components of $\mathbf{e}_{p_L}$ with $\dot{e}_i$ the corresponding velocity error, and e_i, $i = 4, 5, 6$, are the components of $\mathbf{e}_{\Theta_L}$ with $\dot{e}_i$ the corresponding Euler-rate error. Using a two-dimensional feature per channel keeps the number of adaptive parameters at $6 \times 5 = 30$ while avoiding the information collapse of a single shared six-dimensional input.

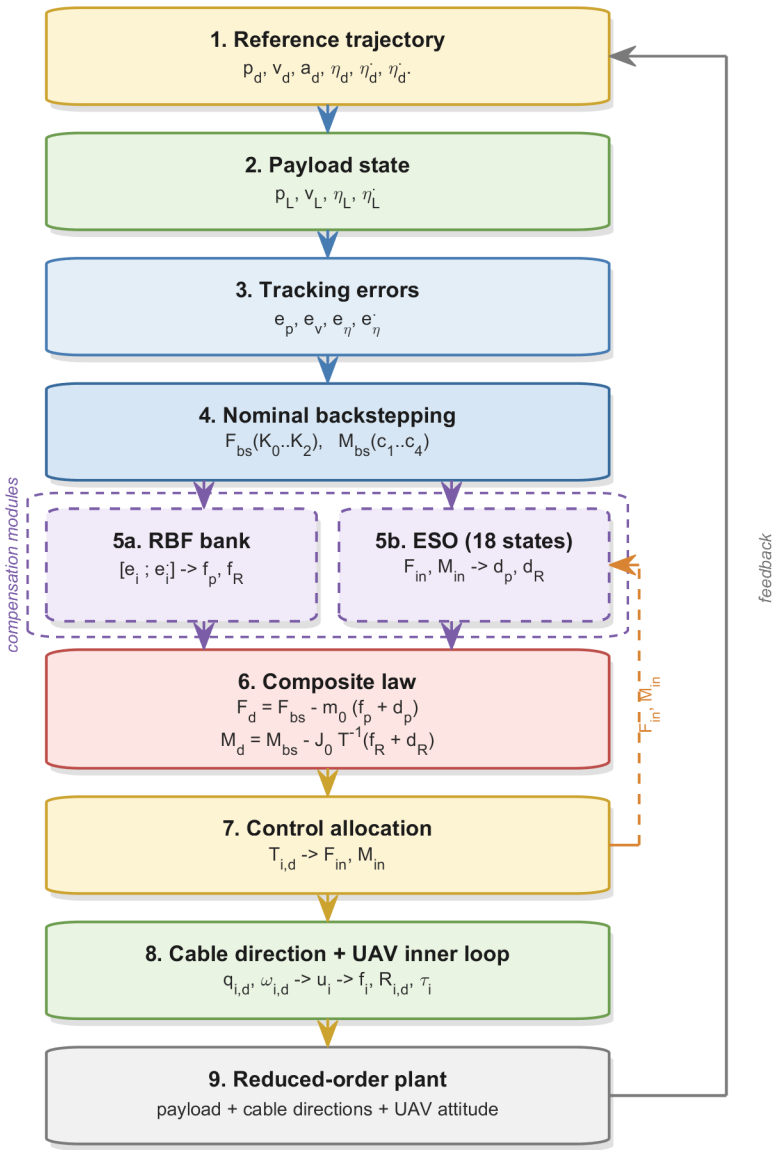


Figure 2. Pipeline of the proposed RBF-ESO backstepping control framework. The ESO prediction model contains the RBF output and is driven by the applied wrench $(\mathbf{F}_{in}, \mathbf{M}_{in})$ returned by the allocation layer.

RBF Neural Network Approximation

RBF networks are widely used for nonlinear function approximation because Gaussian basis functions yield a linearly parameterized approximator with universal approximation capability (Broomhead and Lowe 1988; Park and Sandberg 1991). In this work the approximation is not implemented as a single global network. Instead, six scalar Gaussian RBF networks are used in parallel: three for payload translation and three for payload attitude. Each channel has its own input vector (9), its own basis, and its own adaptive weight vector; the total number of adaptive parameters is therefore $6 \times 5 = 30$.

Before the basis functions are evaluated, each channel is scaled by a fixed diagonal matrix so that both entries of $\boldsymbol{\xi}_i$ have comparable magnitude:

$$\mathbf{x}_i = \mathbf{S}_i^{-1}\boldsymbol{\xi}_i, \qquad \mathbf{S}_i = \begin{cases} \mathrm{diag}(0.5\ \mathrm{m},\ 0.5\ \mathrm{m/s}), & i = 1,2,3, \\ \mathrm{diag}(0.10\ \mathrm{rad},\ 0.50\ \mathrm{rad/s}), & i = 4,5,6. \end{cases} \tag{10}$$

The channel-wise structure is illustrated in Fig. 3.

For the j-th basis node of channel i, the Gaussian activation is

$$h_{i,j} = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{c}_j\|^2}{2b^2}\right), \qquad j = 1,\ldots,5, \tag{11}$$

with a common width $b = 0.8$ and five centres placed at the origin and at the four corners of the normalised plane, $\mathbf{c}_j \in \{(0,0),(\pm 1,\pm 1)\}$. The centres therefore cover the region in which the scaled errors are expected to evolve, and they are not obtained by replicating a scalar into several dimensions. The basis is normalised channel-wise,

$$\bar{\mathbf{h}}_i(\mathbf{x}_i) = \frac{\mathbf{h}_i(\mathbf{x}_i)}{\sum_{j=1}^{5} h_{i,j}(\mathbf{x}_i)}, \qquad \mathbf{h}_i = \begin{bmatrix} h_{i,1} & \cdots & h_{i,5} \end{bmatrix}^T, \tag{12}$$

where the denominator is replaced by a small positive constant when it becomes negligible. The same normalised basis is used for the network output and for the weight update.

For the translational channels, let $\hat{\boldsymbol{\theta}}_{p,k} \in \mathbb{R}^5$, $k = 1,2,3$, denote the adaptive weight vectors of the x-, y-, and z-directions. The RBF outputs are

$$\hat{f}_{p,k} = \hat{\boldsymbol{\theta}}_{p,k}^T \bar{\mathbf{h}}_k, \qquad k = 1,2,3, \tag{13}$$

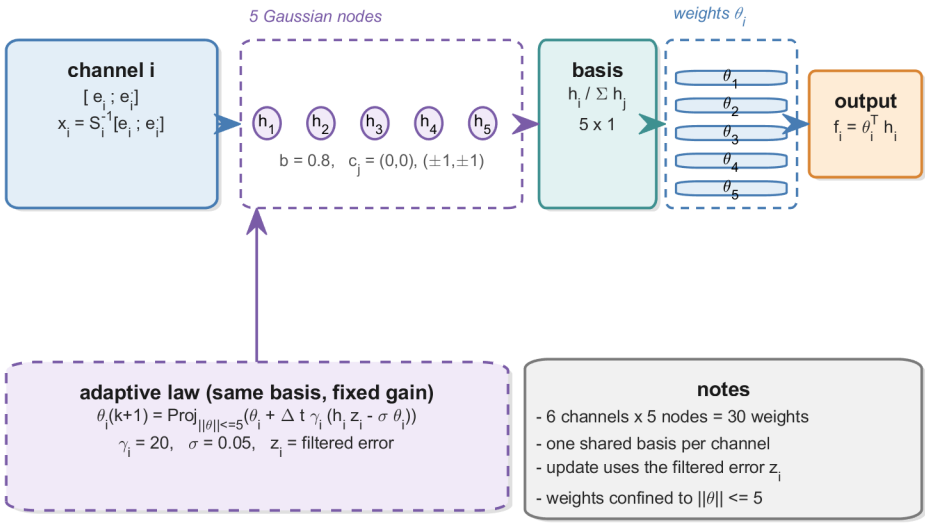


Figure 3. Channel-wise RBF network used for payload uncertainty approximation. One channel is shown; the same five-node structure is repeated for the three translation and three attitude channels, and the output and the adaptation update share the same normalised basis.

stacked as

$$\hat{\mathbf{f}}_p = \begin{bmatrix} \hat{f}_{p,1} & \hat{f}_{p,2} & \hat{f}_{p,3} \end{bmatrix}^T. \tag{14}$$

In the force controller $\hat{\mathbf{f}}_p$ is interpreted as an estimate of acceleration-level uncertainty in the payload translational dynamics, which is why it appears multiplied by the nominal payload mass m_0.

Similarly, for the attitude channels, let $\hat{\boldsymbol{\theta}}_{R,k} \in \mathbb{R}^5$, $k = 1, 2, 3$, denote the weights for roll, pitch, and yaw. The outputs and the stacked vector are

$$\hat{f}_{R,k} = \hat{\boldsymbol{\theta}}_{R,k}^T \bar{\mathbf{h}}_{k+3}, \qquad \hat{\mathbf{f}}_R = \begin{bmatrix} \hat{f}_{R,1} & \hat{f}_{R,2} & \hat{f}_{R,3} \end{bmatrix}^T, \tag{15}$$

which is an estimate of Euler-angle acceleration and is therefore mapped to a moment through the same kinematic matrix as the nominal control, see (31).

The weights are updated online with a channel-wise law that uses the filtered backstepping error of that channel,

$$\dot{\hat{\boldsymbol{\theta}}}_i = \gamma_i \left(\bar{\mathbf{h}}_i z_i - \sigma \hat{\boldsymbol{\theta}}_i \right), \qquad z_i = \begin{cases} e_{v,i} + c_3 e_{p,i}, & i = 1, 2, 3, \\ \dot{e}_{\Theta,i} + c_1 e_{\Theta,i}, & i = 4, 5, 6, \end{cases} \tag{16}$$

with a fixed gain $\gamma_i = 20$ for all six channels and a leakage coefficient $\sigma = 0.05$. The leakage term bounds the weights in the absence of persistent excitation; it is not an error dead zone, and no error-dependent gain scheduling is used. The update is integrated with forward Euler and projected onto the closed ball of radius $\theta_{\max} = 5$,

$$\hat{\boldsymbol{\theta}}_i(k+1) = \mathrm{Proj}_{\|\hat{\boldsymbol{\theta}}\| \le \theta_{\max}} \left(\hat{\boldsymbol{\theta}}_i(k) + \Delta t \, \dot{\hat{\boldsymbol{\theta}}}_i(k) \right). \tag{17}$$

This projection is the only parameter-protection mechanism retained; no element-wise saturation and no output limiting are applied, so the model assumed in the analysis below is the model that is actually implemented.

ESO-Based Residual Estimation

Even with the RBF term active, model residuals and external disturbances remain. Let

$$\mathbf{d}_{\mathrm{res}} = \mathbf{f}_{\mathrm{unk}} - \hat{\mathbf{f}}_{\mathrm{RBF}} + \mathbf{d}_{\mathrm{ext}} \tag{18}$$

be the residual that the network does not explain, where $\mathbf{f}_{\mathrm{unk}}$ collects the unmodeled acceleration-level terms and $\mathbf{d}_{\mathrm{ext}}$ the external disturbance. The ESO estimates $\mathbf{d}_{\mathrm{res}}$, not the external disturbance alone; consequently the RBF parameter error does not enter the analysis twice.

For the translational channel the payload dynamics are written in acceleration form as

$$\ddot{\mathbf{p}}_L = \frac{\mathbf{F}_{in}}{m_0} - g\mathbf{e}_3 + \hat{\mathbf{f}}_p + \mathbf{d}_p, \tag{19}$$

where $\mathbf{F}_{in}$ is the resultant force actually applied by the cables, m_0 the nominal payload mass, and $\mathbf{d}_p$ the translational residual. Note that m_0 appears once: the applied force is divided by the nominal mass, and gravity is grouped with the known terms. Define the augmented state $\mathbf{z}_p = [\mathbf{p}_L^T, \dot{\mathbf{p}}_L^T, \mathbf{d}_p^T]^T$ and the observer

$$\begin{cases} \dot{\hat{\mathbf{z}}}_{p1} = \hat{\mathbf{z}}_{p2} + \boldsymbol{\beta}_{p1} \left(\mathbf{z}_{p1} - \hat{\mathbf{z}}_{p1} \right), \\ \dot{\hat{\mathbf{z}}}_{p2} = \hat{\mathbf{z}}_{p3} + \dfrac{\mathbf{F}_{in}}{m_0} - g\mathbf{e}_3 + \hat{\mathbf{f}}_p + \boldsymbol{\beta}_{p2} \left(\mathbf{z}_{p1} - \hat{\mathbf{z}}_{p1} \right), \\ \dot{\hat{\mathbf{z}}}_{p3} = \boldsymbol{\beta}_{p3} \left(\mathbf{z}_{p1} - \hat{\mathbf{z}}_{p1} \right), \end{cases} \tag{20}$$

where $\mathbf{z}_{p1} = \mathbf{p}_L$, $\mathbf{z}_{p2} = \dot{\mathbf{p}}_L$, $\mathbf{z}_{p3} = \mathbf{d}_p$, and $\hat{\mathbf{z}}_{p3}$ estimates the residual. The gains are parameterised by a single bandwidth $\omega_0 = 50$ rad/s,

$$\boldsymbol{\beta}_{p1} = 3\omega_0 \mathbf{I}_3, \qquad \boldsymbol{\beta}_{p2} = 3\omega_0^2 \mathbf{I}_3, \qquad \boldsymbol{\beta}_{p3} = \omega_0^3 \mathbf{I}_3. \tag{21}$$

For the attitude channel the same construction is applied to the Euler-angle dynamics. Differentiating $\dot{\mathbf{\Theta}}_L = \mathbf{T}\mathbf{\Omega}_L$ gives

$$\begin{aligned} \ddot{\mathbf{\Theta}}_L &= \mathbf{f}_{R,0} + \hat{\mathbf{f}}_R + \mathbf{d}_R, \\ \mathbf{f}_{R,0} &:= \mathbf{T}\mathbf{J}_0^{-1}\left(\mathbf{M}_{in} - \mathbf{\Omega}_L \times \mathbf{J}_0\mathbf{\Omega}_L\right) + \dot{\mathbf{T}}\mathbf{\Omega}_L, \end{aligned} \tag{22}$$

where $\mathbf{M}_{in}$ is the moment actually applied by the cables and $\mathbf{d}_R$ the attitude residual. Note that the inertia is not used as a multiplier of the moment; the Euler-angle acceleration is obtained through $\mathbf{T}\mathbf{J}_0^{-1}(\cdot)$, which is the map implied by the kinematics. The observer has the same structure,

$$\begin{cases} \dot{\hat{\mathbf{z}}}_{\eta 1} = \hat{\mathbf{z}}_{\eta 2} + \boldsymbol{\beta}_{\eta 1}\left(\mathbf{z}_{\eta 1} - \hat{\mathbf{z}}_{\eta 1}\right), \\ \dot{\hat{\mathbf{z}}}_{\eta 2} = \hat{\mathbf{z}}_{\eta 3} + \mathbf{f}_{R,0} + \hat{\mathbf{f}}_R + \boldsymbol{\beta}_{\eta 2}\left(\mathbf{z}_{\eta 1} - \hat{\mathbf{z}}_{\eta 1}\right), \\ \dot{\hat{\mathbf{z}}}_{\eta 3} = \boldsymbol{\beta}_{\eta 3}\left(\mathbf{z}_{\eta 1} - \hat{\mathbf{z}}_{\eta 1}\right), \end{cases} \tag{23}$$

with $\mathbf{z}_{\eta 1} = \mathbf{\Theta}_L$ and the same bandwidth parameterisation. The two blocks give $9 + 9 = 18$ observer states in total.

Because the observer is updated with the wrench of the *previous* interval, the implementation is a discrete-time forward-Euler observer,

$$\hat{\mathbf{z}}(k+1) = \hat{\mathbf{z}}(k) + \Delta t\, f_{\mathrm{ESO}}\left(\hat{\mathbf{z}}(k), \mathbf{F}_{in}(k-1), \mathbf{M}_{in}(k-1), \hat{\mathbf{f}}(k)\right), \tag{24}$$

so that prediction model, measurement, weights, and applied input refer to the same sampling instants.

Payload Control Law

The payload-layer controller computes the desired force $\mathbf{F}_d$ in the inertial frame and the desired moment $\mathbf{M}_d$ in the body frame.

Translational Control With the errors of (8), the virtual control is

$$\boldsymbol{\alpha}_p = -\mathbf{K}_{p1}\mathbf{e}_{p_L} + \dot{\mathbf{p}}_{L,d}, \qquad \mathbf{z}_p = \mathbf{e}_{v_L} + \mathbf{K}_{p1}\mathbf{e}_{p_L}, \tag{25}$$

and the nominal backstepping force is

$$\mathbf{F}_{d,bs} = m_0\left(-g\mathbf{e}_3 + \ddot{\mathbf{p}}_{L,d} - \mathbf{K}_{p1}\mathbf{e}_{v_L} - \mathbf{K}_{p0}\mathbf{e}_{p_L} - \mathbf{K}_{p2}\mathbf{z}_p\right), \tag{26}$$

where $\mathbf{K}_{p1}, \mathbf{K}_{p2} \succ 0$ are diagonal gain matrices. The term $-\mathbf{K}_{p0}\mathbf{e}_{p_L}$ with $\mathbf{K}_{p0} = \mathbf{I}_3$ is the unit proportional gain retained from the original implementation; it is not retuned in this work. Adding the RBF and

residual compensation gives

$$\mathbf{F}_d = \mathbf{F}_{d,bs} - m_0\hat{\mathbf{f}}_p - m_0\hat{\mathbf{z}}_{p3}. \tag{27}$$

Rotational Control With the attitude errors of (8), define

$$\boldsymbol{\alpha}_\eta = -\mathbf{K}_{\eta 1}\mathbf{e}_{\Theta_L} + \dot{\boldsymbol{\Theta}}_{L,d}, \qquad \mathbf{z}_\eta = \mathbf{e}_{\dot{\Theta}_L} + \mathbf{K}_{\eta 1}\mathbf{e}_{\Theta_L}. \tag{28}$$

The nominal Euler-angle acceleration command is

$$\ddot{\boldsymbol{\Theta}}_c = \ddot{\boldsymbol{\Theta}}_{L,d} - \mathbf{K}_{\eta 1}\mathbf{e}_{\dot{\Theta}_L} - \mathbf{e}_{\Theta_L} - \mathbf{K}_{\eta 2}\mathbf{z}_\eta, \tag{29}$$

again with the unit proportional term $-\mathbf{e}_{\Theta_L}$ inherited from the implementation, and the nominal moment follows from the inverse of the kinematic map in (22),

$$\mathbf{M}_{d,bs} = \mathbf{J}_0\mathbf{T}^{-1}\left(\ddot{\boldsymbol{\Theta}}_c - \dot{\mathbf{T}}\boldsymbol{\Omega}_L\right) + \boldsymbol{\Omega}_L \times \mathbf{J}_0\boldsymbol{\Omega}_L. \tag{30}$$

The composite moment is

$$\mathbf{M}_d = \mathbf{M}_{d,bs} - \mathbf{J}_0\mathbf{T}^{-1}\hat{\mathbf{f}}_R - \mathbf{J}_0\mathbf{T}^{-1}\hat{\mathbf{z}}_{\eta 3}, \tag{31}$$

where the inverse kinematic map is required because both $\hat{\mathbf{f}}_R$ and $\hat{\mathbf{z}}_{\eta 3}$ carry the units of Euler-angle acceleration, not of moment.

Stability Analysis

The analysis is deliberately limited to the payload layer: the allocation layer and the quadrotor inner loop enter as bounded implementation errors, which is the honest description of what the reduced-order simulation verifies. The following assumptions are used.

Assumption 1. The desired payload position, attitude, and their first- and second-order derivatives are continuous and bounded.

Assumption 2. On the operating region, the unknown acceleration-level terms of each channel can be represented by constant ideal weights $\boldsymbol{\theta}_i^*$ with a bounded residual ε_i, i.e. $f_i^* = \boldsymbol{\theta}_i^{*T}\bar{\mathbf{h}}_i + \varepsilon_i$ with $|\varepsilon_i| \le \bar{\varepsilon}_i$.

Assumption 3. The normalised basis is bounded, $0 \le \bar{h}_{i,j} \le 1$ and $\sum_j \bar{h}_{i,j} = 1$, and its gradient with respect to the feature vector is bounded on the region reached by the closed loop.

Assumption 4. The residual $\mathbf{d}_{\mathrm{res}}$ of (18) is continuously differentiable and its derivative is bounded on the region reached by the closed loop.

Because the residual contains the learned term, this assumption is checked explicitly below rather than postulated for an external disturbance alone.

Assumption 5. The payload attitude stays in a compact subset on which the ZYX Euler-angle representation, and hence $\mathbf{T}(\boldsymbol{\Theta}_L)$, is non-singular.

Derivative of the learned term. Differentiating $\hat{\mathbf{f}}_i = \hat{\boldsymbol{\theta}}_i^T \bar{\mathbf{h}}_i$ gives

$$\dot{\hat{f}}_i = \dot{\hat{\boldsymbol{\theta}}}_i^T \bar{\mathbf{h}}_i + \hat{\boldsymbol{\theta}}_i^T \frac{\partial \bar{\mathbf{h}}_i}{\partial \mathbf{x}_i} \dot{\mathbf{x}}_i. \tag{32}$$

The first term is bounded because $\bar{\mathbf{h}}_i$ is bounded by Assumption 3 and $\dot{\hat{\boldsymbol{\theta}}}_i$ is bounded by the update (16) together with the projection (44); the second is bounded because the weight vector is confined to the ball of radius $\theta_{\max}$, the basis gradient is bounded by Assumption 3, and $\dot{\mathbf{x}}_i$ is bounded when the tracking errors and the desired trajectory are bounded. Hence $\dot{\hat{f}}_i$ is bounded, and so is $\dot{\mathbf{d}}_{\text{res}}$ in Assumption 4. This is exactly the point at which learning enters the observer error dynamics, and it is not assumed away.

Boundedness of the Observer Errors

Subtracting the observer (20) from the plant (19) gives, per axis,

$$\dot{\tilde{\mathbf{z}}}_{o,p} = \mathbf{A}_p \tilde{\mathbf{z}}_{o,p} + \mathbf{B}_p \dot{\mathbf{d}}_p, \qquad \mathbf{A}_p = \begin{bmatrix} -\boldsymbol{\beta}_{p1}\mathbf{I}_3 & \mathbf{I}_3 & \mathbf{0} \\ -\boldsymbol{\beta}_{p2}\mathbf{I}_3 & \mathbf{0} & \mathbf{I}_3 \\ -\boldsymbol{\beta}_{p3}\mathbf{I}_3 & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \mathbf{B}_p = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{I}_3 \end{bmatrix}, \tag{33}$$

with $\tilde{\mathbf{z}}_{o,p} = [\tilde{\mathbf{z}}_{p1}^T, \tilde{\mathbf{z}}_{p2}^T, \tilde{\mathbf{z}}_{p3}^T]^T$ and $\tilde{\mathbf{z}}_{pj} = \mathbf{z}_{pj} - \hat{\mathbf{z}}_{pj}$. The gains (21) place all eigenvalues of $\mathbf{A}_p$ at $-\omega_0$, so $\mathbf{A}_p$ is Hurwitz and for every $\mathbf{Q}_p \succ 0$ there is a unique $\mathbf{P}_p \succ 0$ with $\mathbf{A}_p^T \mathbf{P}_p + \mathbf{P}_p \mathbf{A}_p = -\mathbf{Q}_p$. With $V_{\text{ESO},p} = \tilde{\mathbf{z}}_{o,p}^T \mathbf{P}_p \tilde{\mathbf{z}}_{o,p}$,

$$\dot{V}_{\text{ESO},p} = -\tilde{\mathbf{z}}_{o,p}^T \mathbf{Q}_p \tilde{\mathbf{z}}_{o,p} + 2\tilde{\mathbf{z}}_{o,p}^T \mathbf{P}_p \mathbf{B}_p \dot{\mathbf{d}}_p \le -c_{p1} \|\tilde{\mathbf{z}}_{o,p}\|^2 + c_{p2}, \tag{34}$$

by Young's inequality and Assumption 4, with $c_{p1} > 0$ whenever $\lambda_{\min}(\mathbf{Q}_p) > 2\|\mathbf{P}_p \mathbf{B}_p\| \bar{d}'_p$ and c_{p2} proportional to $(\|\mathbf{P}_p \mathbf{B}_p\| \bar{d}'_p)^2$. The attitude block is identical with $\mathbf{A}_\eta$, $\mathbf{P}_\eta$, and $\tilde{\mathbf{z}}_{o,\eta}$, so both observer errors are uniformly ultimately bounded; they converge exponentially to the origin when the residual derivative vanishes.

Closed-Loop Boundedness

Substituting (27) and (31) into the payload dynamics and using the nominal model gives, with $\boldsymbol{\chi}_p = [\mathbf{e}_{p_L}^T, \mathbf{z}_p^T]^T$ and $\boldsymbol{\chi}_\eta = [\mathbf{e}_{\Theta_L}^T, \mathbf{z}_\eta^T]^T$,

$$\begin{aligned} \dot{\boldsymbol{\chi}}_p &= \mathbf{A}_{c,p}\boldsymbol{\chi}_p + \mathbf{B}_c\left(-\tilde{\boldsymbol{\Theta}}_p^T\bar{\mathbf{h}} + \boldsymbol{\Delta}_p\right), \\ \dot{\boldsymbol{\chi}}_\eta &= \mathbf{A}_{c,\eta}\boldsymbol{\chi}_\eta + \mathbf{B}_c\left(-\tilde{\boldsymbol{\Theta}}_R^T\bar{\mathbf{h}} + \boldsymbol{\Delta}_\eta\right), \end{aligned} \tag{35}$$

where $\bar{\mathbf{h}}$ stacks the channel bases, $\tilde{\boldsymbol{\Theta}}_p = [\tilde{\boldsymbol{\theta}}_{p,1}, \tilde{\boldsymbol{\theta}}_{p,2}, \tilde{\boldsymbol{\theta}}_{p,3}]$, $\tilde{\boldsymbol{\theta}}_i = \hat{\boldsymbol{\theta}}_i - \boldsymbol{\theta}_i^*$, $\boldsymbol{\Delta}_p = \boldsymbol{\varepsilon}_p + \tilde{\mathbf{z}}_{p3}$ and $\boldsymbol{\Delta}_\eta = \boldsymbol{\varepsilon}_\eta + \tilde{\mathbf{z}}_{\eta 3}$ collect the bounded approximation and observer residuals, and

$$\mathbf{A}_{c,p} = \begin{bmatrix} -\mathbf{K}_{p1} & \mathbf{I}_3 \\ -\mathbf{K}_{p0} & -\mathbf{K}_{p2} \end{bmatrix}, \quad \mathbf{A}_{c,\eta} = \begin{bmatrix} -\mathbf{K}_{\eta 1} & \mathbf{I}_3 \\ -\mathbf{I}_3 & -\mathbf{K}_{\eta 2} \end{bmatrix}, \quad \mathbf{B}_c = \begin{bmatrix} \mathbf{0} \\ \mathbf{I}_3 \end{bmatrix}. \tag{36}$$

The lower-left blocks are not zero: they come from the proportional terms retained in (26) and (29). With $\mathbf{K}_{p0}, \mathbf{K}_{p1}, \mathbf{K}_{p2}, \mathbf{K}_{\eta 1}, \mathbf{K}_{\eta 2} \succ 0$ both $\mathbf{A}_{c,p}$ and $\mathbf{A}_{c,\eta}$ are Hurwitz, so for any $\mathbf{Q}_1, \mathbf{Q}_2 \succ 0$ there exist $\mathbf{P}_1, \mathbf{P}_2 \succ 0$ solving $\mathbf{A}_{c,p}^T\mathbf{P}_1 + \mathbf{P}_1\mathbf{A}_{c,p} = -\mathbf{Q}_1$ and $\mathbf{A}_{c,\eta}^T\mathbf{P}_2 + \mathbf{P}_2\mathbf{A}_{c,\eta} = -\mathbf{Q}_2$.

Take the Lyapunov function

$$V = \frac{1}{2}\boldsymbol{\chi}_p^T\mathbf{P}_1\boldsymbol{\chi}_p + \frac{1}{2}\boldsymbol{\chi}_\eta^T\mathbf{P}_2\boldsymbol{\chi}_\eta + \frac{1}{2}\sum_{i=1}^{6}\tilde{\boldsymbol{\theta}}_i^T\boldsymbol{\Gamma}_i^{-1}\tilde{\boldsymbol{\theta}}_i, \qquad \boldsymbol{\Gamma}_i = \gamma_i\mathbf{I}_5. \tag{37}$$

Because the feature vector of each channel is used in that channel only, the cross term produced by $\mathbf{B}_c$ separates channel-wise:

$$\boldsymbol{\chi}^T\mathbf{P}\mathbf{B}_c\left(-\tilde{\boldsymbol{\Theta}}^T\bar{\mathbf{h}}\right) = -\sum_{i=1}^{6}\left(\mathbf{p}_{2,i}^T\mathbf{z}\right)\tilde{\boldsymbol{\theta}}_i^T\bar{\mathbf{h}}_i, \tag{38}$$

where $\mathbf{p}_{2,i}$ denotes the block of $\mathbf{P}$ acting on $\mathbf{z}$. Choosing the channel gains such that $\gamma_i\mathbf{p}_{2,i} = \mathbf{e}_i$, the adaptation term of (16) cancels (38) exactly; if the gains are fixed in advance and the identity does not hold, the mismatch is bounded and is absorbed into the constant c below. Differentiating V and using Young's inequality on the residual and leakage terms yields

$$\dot{V} \le -\lambda_1\|\boldsymbol{\chi}_p\|^2 - \lambda_2\|\boldsymbol{\chi}_\eta\|^2 - \lambda_3\|\tilde{\boldsymbol{\Theta}}_p\|_F^2 - \lambda_4\|\tilde{\boldsymbol{\Theta}}_R\|_F^2 + c, \tag{39}$$

with $\lambda_{1,2} > 0$ provided $\lambda_{\min}(\mathbf{Q}_{1,2})/2 > \delta_{p,\eta}\|\mathbf{P}_{1,2}\mathbf{B}_c\|$ and $\lambda_{3,4} = \sigma/2$ from the leakage term. The constant $c > 0$ collects the approximation

residuals $\bar{\varepsilon}$, the observer residuals $\bar{\Delta}$, the leakage bias $\sigma\|\boldsymbol{\theta}^*\|^2/2$, and the bounded mismatch of the gain choice above. Since V is quadratically bounded by $\mathbf{y} = [\boldsymbol{\chi}_p^T, \boldsymbol{\chi}_\eta^T, \tilde{\boldsymbol{\theta}}_1^T, \ldots, \tilde{\boldsymbol{\theta}}_6^T]^T$, there exist $\underline{\alpha}, \overline{\alpha} > 0$ with $\underline{\alpha}\|\mathbf{y}\|^2 \le V \le \overline{\alpha}\|\mathbf{y}\|^2$, and (39) implies

$$V(t) \le V(0)e^{-\alpha t} + \frac{\beta}{\alpha}\left(1 - e^{-\alpha t}\right), \qquad \alpha = \frac{\min(\lambda_1, \ldots, \lambda_4)}{\overline{\alpha}}, \quad \beta = \frac{c}{\underline{\alpha}}, \tag{40}$$

so all components of $\mathbf{y}$ are uniformly ultimately bounded with ultimate radius proportional to $\sqrt{\beta/\alpha}$.

The conclusion is conditional and it is stated as such: under Assumptions 1–5, the payload position error, the attitude error, the two backstepping filtered errors, the RBF weights, and the observer errors remain bounded; the bound degrades with the approximation residual, the residual derivative, and the leakage bias. The analysis does not establish asymptotic convergence unless those residuals vanish, it does not cover the allocation or inner-loop dynamics except as bounded disturbances, and it is a continuous-time statement whose discrete implementation is a separate forward-Euler approximation.

Remark The two modules therefore play asymmetric roles in the analysis as well as in the data. The observer absorbs the dominant part of the residual; the network contributes a bounded correction whose effect on the closed loop is captured by the projection radius and the leakage bias, not by a separable share of the disturbance.

Control Allocation and Cable Force Generation

Following the geometric cooperative transport formulation in Lee et al. (2013a), the payload-level controller first computes the desired wrench for payload motion tracking, consisting of a resultant force $\mathbf{F}_d \in \mathbb{R}^3$ expressed in the inertial frame and a resultant moment $\mathbf{M}_d \in \mathbb{R}^3$ expressed in the payload body frame. This pair provides a compact interface between high-level tracking control and low-level actuation. Since the payload is driven through multiple cables rather than a single actuator, the allocation module maps the desired wrench to individual cable forces while preserving the overall force–moment balance required by the payload dynamics.

The desired resultant force and moment on the payload are

$$\mathbf{F}_d = \sum_{i=1}^{n} \mathbf{T}_{i,d}, \qquad \mathbf{M}_d = \sum_{i=1}^{n} \hat{\boldsymbol{\rho}}_i \mathbf{R}_L^T \mathbf{T}_{i,d}, \tag{41}$$

where $\mathbf{T}_{i,d}$ is the desired force of the i-th cable and $\hat{\boldsymbol{\rho}}_i = S(\boldsymbol{\rho}_i)$ is the skew-symmetric matrix of the attachment point. Let

$$\mathbf{P} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{I}_3 & \cdots & \mathbf{I}_3 \\ \hat{\boldsymbol{\rho}}_1 & \hat{\boldsymbol{\rho}}_2 & \cdots & \hat{\boldsymbol{\rho}}_n \end{bmatrix}, \qquad \mathbf{w}_d = \begin{bmatrix} \mathbf{R}_L^T \mathbf{F}_d \\ \mathbf{M}_d \end{bmatrix}, \tag{42}$$

so that the minimum-norm cable forces in body coordinates are

$$\mathbf{T}_d^b = \mathbf{P}^T \left(\mathbf{P}\mathbf{P}^T\right)^{-1} \mathbf{w}_d, \qquad \mathbf{T}_d = \operatorname{diag}\left(\mathbf{R}_L, \ldots, \mathbf{R}_L\right) \mathbf{T}_d^b, \tag{43}$$

with the Moore–Penrose pseudoinverse used when $\mathbf{P}\mathbf{P}^T$ is singular.

Because each cable can only pull along its current direction $\mathbf{q}_i$, the realized force is the projection

$$\boldsymbol{\mu}_i = \mathbf{q}_i \mathbf{q}_i^T \mathbf{T}_{i,d}, \qquad \mathbf{F}_{in} = \sum_{i=1}^{n} \boldsymbol{\mu}_i, \qquad \mathbf{M}_{in} = \sum_{i=1}^{n} \hat{\boldsymbol{\rho}}_i \mathbf{R}_L^T \boldsymbol{\mu}_i. \tag{44}$$

The applied wrench $(\mathbf{F}_{in}, \mathbf{M}_{in})$ is what the residual observer described above uses, and the difference between commanded and applied wrench is logged as the allocation residual.

Two remarks are necessary. First, with three cables the allocation has only three scalar degrees of freedom and cannot realise an arbitrary six-dimensional wrench; the pseudoinverse above distributes the residual in a least-squares sense, so an allocation error is expected and is reported. Second, the pseudoinverse does not guarantee positive tension. The signed indicator used in this paper is

$$T_i^{\text{model}} = -\mathbf{q}_i^T \boldsymbol{\mu}_i, \tag{45}$$

which is a property of the reduced model, not a physical cable tension: it is reported for every run so that a loss of tautness would be visible, and no claim of permanent tautness is made.

Cable Direction Tracking and Quadrotor Force Command

Following Lee et al. (2013a), the desired cable direction is defined from the allocated force,

$$\mathbf{q}_{i,d} = -\frac{\mathbf{T}_{i,d}}{\|\mathbf{T}_{i,d}\|}, \qquad \boldsymbol{\omega}_{i,d} = \mathbf{q}_{i,d} \times \dot{\mathbf{q}}_{i,d}, \tag{46}$$

and the cable tracking errors are

$$\mathbf{e}_{q,i} = \mathbf{q}_{i,d} \times \mathbf{q}_i, \qquad \mathbf{e}_{\omega,i} = \boldsymbol{\omega}_i + \hat{\mathbf{q}}_i^2 \boldsymbol{\omega}_{i,d}. \tag{47}$$

The feedforward term that accounts for payload translation and rotation is

$$\mathbf{a}_i = \frac{1}{m_L}\mathbf{F}_{in} + \mathbf{R}_L \hat{\boldsymbol{\Omega}}_L^2 \boldsymbol{\rho}_i + \mathbf{R}_L \hat{\boldsymbol{\rho}}_i \mathbf{J}_L^{-1} \left(\hat{\boldsymbol{\Omega}}_L \mathbf{J}_L \boldsymbol{\Omega}_L - \mathbf{M}_{in} \right), \tag{48}$$

and the commanded force of the i-th quadrotor is decomposed as $\mathbf{u}_i = \mathbf{u}_{i,\perp} + \mathbf{u}_{i,\parallel}$ with

$$\begin{aligned} \mathbf{u}_{i,\perp} &= m_{p,i} l_i \hat{\mathbf{q}}_i \Big(-\mathbf{K}_{q,i}\mathbf{e}_{q,i} - \mathbf{K}_{\omega,i}\mathbf{e}_{\omega,i} - \left(\mathbf{q}_i^T \boldsymbol{\omega}_{i,d}\right) \dot{\mathbf{q}}_i \\ &\qquad - \hat{\mathbf{q}}_i^2 \dot{\boldsymbol{\omega}}_{i,d} \Big) - m_{p,i} \hat{\mathbf{q}}_i^2 \mathbf{a}_i, \\ \mathbf{u}_{i,\parallel} &= \mathbf{T}_{i,d} + m_{p,i} \left(l_i \|\boldsymbol{\omega}_i\|^2 \mathbf{q}_i + \mathbf{q}_i \mathbf{q}_i^T \mathbf{a}_i \right). \end{aligned} \tag{49}$$

In the implementation the cable direction is integrated once per step and renormalised; no ad hoc interpolation of the realised direction toward the desired one is used, because such a correction would hide a modelling inconsistency.

Quadrotor Thrust and Attitude Command

Following the geometric thrust-direction construction in Lee et al. (2013a,b), the cable-level controller provides each quadrotor with a commanded force vector $\mathbf{u}_i \in \mathbb{R}^3$ in the inertial frame. The desired body-fixed thrust direction aligns with that force,

$$\mathbf{b}_{3,i,d} = \frac{\mathbf{u}_i}{\|\mathbf{u}_i\|}, \qquad \mathbf{b}_{1,\psi,i} = \begin{bmatrix} \cos\psi_{i,d} \\ \sin\psi_{i,d} \\ 0 \end{bmatrix}, \tag{50}$$

so that an orthonormal desired frame is

$$\begin{aligned} \mathbf{b}_{2,i,d} &= \frac{\mathbf{b}_{3,i,d} \times \mathbf{b}_{1,\psi,i}}{\|\mathbf{b}_{3,i,d} \times \mathbf{b}_{1,\psi,i}\|}, \qquad \mathbf{b}_{1,i,d} = \mathbf{b}_{2,i,d} \times \mathbf{b}_{3,i,d}, \\ \mathbf{R}_{i,d} &= \begin{bmatrix} \mathbf{b}_{1,i,d} & \mathbf{b}_{2,i,d} & \mathbf{b}_{3,i,d} \end{bmatrix}. \end{aligned} \tag{51}$$

The thrust magnitude is the projection of the commanded force on the current body z-axis,

$$f_i = \mathbf{u}_i^T \mathbf{R}_i \mathbf{e}_3. \tag{52}$$

The attitude of each quadrotor is tracked by the same backstepping structure as the payload, in Euler-angle coordinates,

$$\boldsymbol{\eta}_i = [\phi_i, \theta_i, \psi_i]^T, \qquad \mathbf{z}_{1,i} = \boldsymbol{\eta}_i - \boldsymbol{\eta}_{i,d}, \qquad \mathbf{z}_{2,i} = \dot{\boldsymbol{\eta}}_i - \dot{\boldsymbol{\eta}}_{i,d} + \mathbf{C}_{1,i}\mathbf{z}_{1,i}, \tag{53}$$

$$\boldsymbol{\tau}_i = \mathbf{B}_i^{-1} \left(-\mathbf{z}_{1,i} - \mathbf{c}_i(\dot{\boldsymbol{\eta}}_i) + \dot{\boldsymbol{\alpha}}_{1,i} - \mathbf{C}_{2,i}\mathbf{z}_{2,i} \right), \tag{54}$$

where $\mathbf{B}_i$ is the input gain matrix, $\mathbf{C}_{1,i}, \mathbf{C}_{2,i} \succ 0$ are gain matrices, and $\mathbf{c}_i(\dot{\boldsymbol{\eta}}_i)$ is the Coriolis/gyroscopic term. The inner loop is not the subject of this paper; it is documented here so that the reported payload-level results can be reproduced.

Simulation Setup

The evaluation uses a hierarchical reduced-order simulation in MATLAB. The payload is integrated as a 6-DOF rigid body, the payload-level controller produces a desired wrench, the allocation layer maps it to cable forces, and each quadrotor tracks the resulting thrust direction with an Euler-angle inner loop. The payload receives the force that the cables actually apply; the quadrotor states are simulated but are not fed back into the payload through a physical cable model. The model is therefore a reduced-order, control-oriented representation of the cooperative system, and the results below should be read at that level. It does not include cable slackening, actuator saturation, or aerodynamic interaction between the vehicles.

Physical Parameters and Numerical Integration

The dynamics are integrated with a fixed-step fourth-order Runge–Kutta scheme using $\Delta t = 0.001$ s and a horizon of $T_f = 20$ s, i.e. 20001 samples and 20000 integration intervals. The principal parameters are listed in the Appendix. Unless stated otherwise, the three

quadrotors are identical, each with mass $m_p = 2.0$ kg and inertia $\mathbf{J}_p = \mathrm{diag}(0.082, 0.0845, 0.1377)$ kg·m^2; the payload has $m_L = 2.0$ kg, dimensions $1.0 \times 0.8 \times 0.2$ m, and $\mathbf{J}_L = \mathrm{diag}(0.1133, 0.1733, 0.2733)$ kg·m^2.

Initial Conditions and Reference Trajectory

The quadrotors start from a near-hover configuration with zero attitude angles, zero angular rates, and zero translational velocities, at

$$\begin{aligned} \mathbf{p}_1(0) = [0.5,\, 4.2,\, 0]^T, \quad \mathbf{p}_2(0) = [-0.5,\, 4.6,\, 0]^T, \\ \mathbf{p}_3(0) = [-0.5,\, 3.8,\, 0]^T, \end{aligned} \tag{55}$$

which is slightly asymmetric and therefore does not produce a perfectly symmetric transient. The payload starts at $\mathbf{p}_L(0) = [0,\, 4.2,\, 0.1]^T$ with zero velocity and zero angular velocity, and the attachment points in the body frame are

$$\begin{aligned} \boldsymbol{\rho}_1 = [0.5,\, 0.0,\, 0.1]^T, \qquad \boldsymbol{\rho}_2 = [-0.5,\, 0.4,\, 0.1]^T, \\ \boldsymbol{\rho}_3 = [-0.5,\, -0.4,\, 0.1]^T. \end{aligned} \tag{56}$$

The simulation starts directly in the transport phase; no takeoff phase is implemented. The desired payload position is

$$\mathbf{p}_{L,d}(t) = \begin{bmatrix} 1.5\sin(\omega_d t) \\ 4.2\cos(\omega_d t/2) \\ 0.5 \end{bmatrix}, \qquad \omega_d = \frac{2\pi}{10}\ \mathrm{rad/s}, \tag{57}$$

with analytic velocity and acceleration references, and the desired attitude follows the tangent direction of the path,

$$\begin{aligned} \mathbf{r}_{d1} &= \frac{\dot{\mathbf{p}}_{L,d}}{\|\dot{\mathbf{p}}_{L,d}\|}, \qquad \mathbf{r}_{d3} = \mathbf{e}_3, \qquad \mathbf{r}_{d2} = \frac{\mathbf{r}_{d3} \times \mathbf{r}_{d1}}{\|\mathbf{r}_{d3} \times \mathbf{r}_{d1}\|}, \\ \mathbf{R}_{L,d} &= \begin{bmatrix} \mathbf{r}_{d1} & \mathbf{r}_{d2} & \mathbf{r}_{d3} \end{bmatrix}, \end{aligned} \tag{58}$$

with $\mathbf{R}_{L,d} = \mathbf{I}_3$ when the reference speed is negligible.

Disturbance Injection

Two disturbance classes are used. The periodic one is the multi-harmonic signal of the submitted version,

$$\begin{aligned} d_p(t) &= 3\sin t + 0.2\sin 2t + 0.3\sin 3t + 0.4\sin 4t, \\ d_R(t) &= 1\sin t + 0.2\sin 2t + 0.3\sin 3t + 0.4\sin 4t, \end{aligned} \tag{59}$$

injected with the same scalar in each translational and each rotational channel. The non-periodic one is a band-limited random signal generated as first-order low-pass filtered white noise,

$$x_k = a x_{k-1} + \sqrt{1-a^2}\, w_k, \qquad a = e^{-2\pi f_c \Delta t}, \qquad \mathbf{x} \leftarrow \frac{\mathbf{x} - \overline{\mathbf{x}}}{\sqrt{\overline{\mathbf{x}^2}}}\, \mathbf{A}, \tag{60}$$

with $f_c = 1$ Hz, translational RMS $A_p = 1.0$ m/s^2, attitude RMS $A_R = 0.3$ rad/s^2, and three independent channels in each block. Each seed initialises the generator once, and the trace is stored with a hash so that a run can be reproduced exactly.

The two classes are mutually exclusive: when the random disturbance is enabled, the periodic signal of (59) is not added, so the external excitation is never counted twice. When the random disturbance is disabled, the injection path is bitwise identical to the submitted code, which was verified by comparing the state trajectories of a reference run before and after the modification.

Metrics, Failure Criterion, and Protocol

All metrics are evaluated on the final 20% of the time window, $t \geq 0.8 T_f$, and the full-window values are reported as a cross-check. The following quantities are used: the payload position RMS $\sqrt{\mathrm{mean}(\|\mathbf{e}_{p_L}\|^2)}$ and the maximum position error; the rotation-angle error

$$\theta(t) = \arccos\left(\mathrm{clip}\left(\frac{\mathrm{tr}(\mathbf{R}_{L,d}^T \mathbf{R}_L) - 1}{2}, -1, 1\right)\right), \tag{61}$$

reported as RMS and maximum; the signed tension indicator of (45) and the number of negative samples; the allocation force residual; and the number of failed runs. A run is declared failed, and kept in the table, if any metric is non-finite, if the position error exceeds 5 m, or if the integration does not reach T_f.

The protocol freezes one common parameter set for all four configurations (Table 7); only the compensation switches change. The matrix is: four configurations under no disturbance and under the periodic disturbance (8 runs); the mass/inertia mismatch sweep at $0.5, 0.8, 1.0, 1.2, 1.5$ times the nominal values with the periodic disturbance (18 runs); the unmodeled drag levels $c_d = 0, 4, 12$ with the

Table 1. Backstepping-family ablation cases.

Configuration	RBF compensation	ESO compensation
Backstepping	Off	Off
Backstepping+RBF	On	Off
Backstepping+ESO	Off	On
Backstepping+RBF+ESO	On	On

periodic disturbance (12 runs); and the non-periodic disturbance with five fixed seeds (20 runs). The mismatch sweep multiplies mass and inertia together, so it probes a combined mismatch and does not separate the two effects.

Simulation Results and Discussion

The results are organised as a reader-facing sequence of questions: whether the ordering of the configurations survives a change of disturbance class, what the observer contributes on its own, and how much the network adds on top of it.

Backstepping-Family Ablation

Table 2 reports the ablation, and Figs. 4 and 5 show the corresponding time histories. Under no disturbance all four configurations track the reference to the numerical floor: the position error is between 9×10^{-6} and 9×10^{-5} m, which is the level of the integrator and not a property of any module. This condition therefore carries no discriminative information, a point that matters when reading the figures.

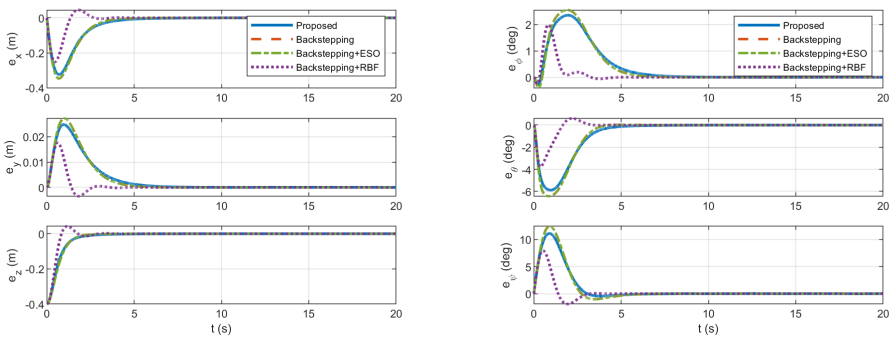
Under the periodic disturbance the ordering is unambiguous: plain backstepping is the worst (3.38×10^{-1} m), the ESO-only configuration reduces the error by a factor of about thirteen, and adding the RBF term reduces it by a further 9.7%. The same ordering holds for the rotation-angle error and for the maximum error. All runs completed and no negative tension sample was recorded; the minimum signed tension over the ablation is 3.62 N, well inside the taut regime of the reduced model.

Parameter Mismatch and Unmodeled Drag

Tables 3 and 4 report the two robustness probes that use the periodic disturbance. The nominal controller is kept unchanged in both cases: the plant mass and inertia are scaled, or a quadratic drag force $-\,c_d\|\mathbf{v}\|\mathbf{v}$ is

Table 2. Backstepping-family ablation: payload position RMS and maximum, rotation-angle RMS, and minimum signed tension, on the final 20% of the window. The last column is the ratio of the composite to the ESO-only position RMS. No negative tension sample occurred in any run.

Configuration	Position RMS (m)	Position max (m)	Rotation RMS (rad)	$T_{\min}$ (N)	Ratio
No disturbance					
Backstepping	3.08×10^{-5}	4.55×10^{-5}	7.09×10^{-5}	4.762	–
Backstepping+RBF	9.22×10^{-6}	1.44×10^{-5}	4.15×10^{-5}	4.762	–
Backstepping+ESO	9.25×10^{-5}	1.40×10^{-4}	1.31×10^{-4}	4.762	–
Backstepping+RBF+ESO	8.68×10^{-5}	1.33×10^{-4}	1.17×10^{-4}	4.762	–
Periodic disturbance					
Backstepping	3.38×10^{-1}	4.79×10^{-1}	1.21×10^{-1}	3.621	–
Backstepping+RBF	1.62×10^{-1}	2.66×10^{-1}	7.07×10^{-2}	3.800	–
Backstepping+ESO	2.62×10^{-2}	4.61×10^{-2}	1.15×10^{-2}	3.634	1.000
Backstepping+RBF+ESO	2.36×10^{-2}	4.16×10^{-2}	1.06×10^{-2}	3.632	0.903

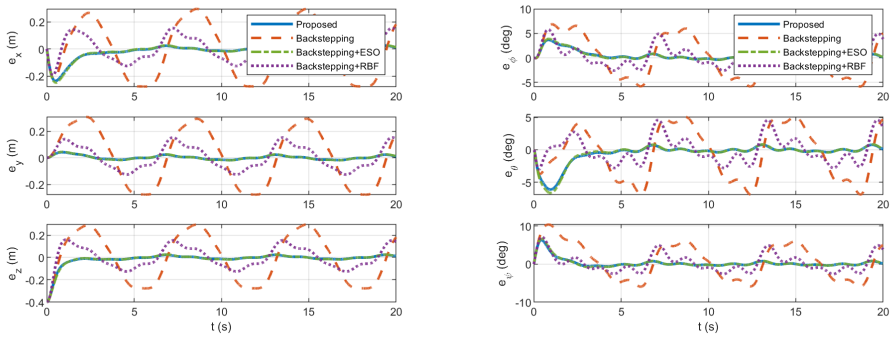


(a) Position error (b) Attitude error

Figure 4. Backstepping-family ablation without disturbance injection. All four configurations reach the numerical floor, so this condition is shown for completeness rather than as evidence of a module.

added to the payload, which reaches about 74% of the payload weight at $c_d = 12$.

The ESO-only configuration absorbs the mismatch: the error grows only from 1.27×10^{-2} to 3.99×10^{-2} m while the plant is scaled from 0.5 to 1.5 times the nominal value, a factor of three in the plant against a factor of about three in the error. The composite-to-ESO ratio, however, stays at 0.903 across the whole sweep, and it also stays at 0.903–0.907 as the drag grows. A module that learned the mismatch or the velocity-dependent drag would show a ratio that decreases with the severity of the perturbation; the ratio does not move. The RBF term behaves instead like a roughly constant scaling of the residual error.



(a) Position error (b) Attitude error

Figure 5. Backstepping-family ablation with the periodic disturbance. The observer removes most of the periodic error; the network adds a further modest reduction.

Table 3. Payload position RMS (m) under mass/inertia mismatch, periodic disturbance. The last column is the composite-to-ESO ratio. Cells marked "–" were not run, because the plain backstepping case diverges far outside the operating range of the comparison.

Scale	BS	BS+RBF	BS+ESO	BS+RBF+ESO	Ratio
0.5	–	1.91×10^{-1}	1.27×10^{-2}	1.14×10^{-2}	0.903
0.8	2.68×10^{-1}	1.29×10^{-1}	2.07×10^{-2}	1.87×10^{-2}	0.903
1.0	3.38×10^{-1}	1.62×10^{-1}	2.62×10^{-2}	2.36×10^{-2}	0.903
1.2	5.39×10^{-1}	2.49×10^{-1}	3.16×10^{-2}	2.86×10^{-2}	0.903
1.5	–	5.26×10^{-1}	3.99×10^{-2}	3.60×10^{-2}	0.903

Table 4. Payload position RMS (m) with an unmodeled quadratic drag, periodic disturbance. The drag coefficient c_d has units $\mathrm{N\,s^2/m^2}$.

c_d	BS	BS+RBF	BS+ESO	BS+RBF+ESO	Ratio
0	3.38×10^{-1}	1.62×10^{-1}	2.62×10^{-2}	2.36×10^{-2}	0.903
4	4.84×10^{-1}	2.13×10^{-1}	2.82×10^{-2}	2.55×10^{-2}	0.904
12	9.85×10^{-1}	6.57×10^{-1}	4.08×10^{-2}	3.70×10^{-2}	0.907

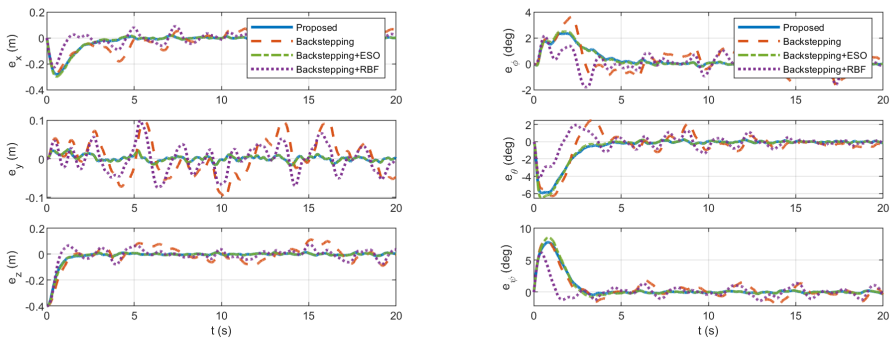
Non-Periodic Disturbance

The random-disturbance experiment replaces the periodic signal by (60) and repeats the four configurations on five fixed seeds. All twenty runs completed, no metric was non-finite, no run exceeded the 5 m failure threshold, and no negative tension sample was recorded.

The ordering is the same as in the periodic case, and it is the same on every seed rather than only on average: for each of the five seeds the

Table 5. Non-periodic disturbance, five seeds: mean $\pm$ standard deviation over the seeds, with the worst-case minimum tension and the total number of failed runs.

Configuration	Position RMS (m)	Position max (m)	Rotation RMS (rad)	Tension min (N)
Backstepping	$8.77 \times 10^{-2} \pm 1.53 \times 10^{-2}$	1.41×10^{-1}	$2.47 \times 10^{-2} \pm 7.33 \times 10^{-3}$	3.518
Backstepping+RBF	$6.28 \times 10^{-2} \pm 1.05 \times 10^{-2}$	1.04×10^{-1}	$1.74 \times 10^{-2} \pm 3.24 \times 10^{-3}$	3.319
Backstepping+ESO	$1.15 \times 10^{-2} \pm 1.58 \times 10^{-3}$	1.88×10^{-2}	$3.39 \times 10^{-3} \pm 4.59 \times 10^{-4}$	3.213
Backstepping+RBF+ESO	$1.08 \times 10^{-2} \pm 1.48 \times 10^{-3}$	1.78×10^{-2}	$3.21 \times 10^{-3} \pm 4.24 \times 10^{-4}$	3.204



(a) Position error

(b) Attitude error

Figure 6. Non-periodic band-limited disturbance, seed 101. The broadband excitation produces a less regular error pattern than the periodic case, but the ordering of the four configurations is unchanged.

composite is better than ESO-only, ESO-only is better than RBF-only, and RBF-only is better than plain backstepping. The magnitude of the RBF contribution shrinks: the composite-to-ESO ratio is 0.943 ± 0.011, i.e. about 5.8%, against 0.903 under the periodic signal, while the observer still reduces the error relative to plain backstepping by a factor of about seven.

These numbers must be compared with care. The two disturbance classes do not have the same RMS: the periodic signal of (59) has an RMS of about 2.15 m/s^2 in translation, whereas the random one is normalised to 1.0 m/s^2. The absolute errors of the random runs are therefore lower for all configurations, and the comparison is meaningful for the ordering and for the ratio, not for absolute accuracy. Five seeds provide an indication of dispersion, not statistical significance.

Interpretation

Three statements are supported by the data. First, the observer is the mechanism behind the large reduction: it lowers the payload error by an

order of magnitude under the periodic disturbance, by about seven times under the random one, and it keeps the error within a factor of three when the plant is scaled over a factor-of-three range. Second, the network contributes a consistent but modest margin, 5.8–9.7% depending on the disturbance class, and the margin does not grow when the mismatch or the unmodeled nonlinearity becomes more severe. Third, the ordering of the four configurations is invariant across the disturbance classes tested, which is what makes the ablation a usable comparison rather than an artefact of one disturbance shape.

What the data do not support is equally important. They do not show that the network identifies the unmodeled dynamics: a five-node static map of the tracking error cannot represent a velocity-dependent drag, and the weight norms remain small when the observer is active. They do not establish a division of labour between learning and observation, and they do not establish that the composite is the best choice in every condition; under no disturbance the ESO-only configuration is in fact slightly worse than plain backstepping, which is a numerical-floor effect rather than a performance difference. Finally, five seeds and a single trajectory are not a statistical study.

Limitations

The evidence is bounded in several ways that the reader should weigh together. The plant is a hierarchical reduced-order model: the payload is driven by the allocated cable forces, but cable slackening, tension bounds, actuator saturation, and aerodynamic interaction are not represented, so the reported tensions are indicators of the reduced model rather than physical cable tensions. The stability statement is conditional and covers the payload layer with bounded implementation errors; it does not claim a full-system guarantee. No hardware or hardware-in-the-loop experiment was performed. The comparison is internal to the backstepping family: an external baseline such as a sliding-mode or a non-neural adaptive backstepping controller would be needed before claiming an advantage over the literature, and this revision does not provide one. Computational cost is not reported here either; the controller is evaluated in MATLAB on a desktop machine, which would not support a hard real-time claim. Finally, the mismatch sweep scales mass and inertia together,

and the random experiment uses five seeds, so neither separates individual parameter effects nor supports significance statements.

Conclusion

This paper presented a payload-level RBF-ESO backstepping framework in which the network output enters the observer prediction model, so that the observer estimates the residual that remains after learning. The framework was evaluated with one frozen parameter set in a backstepping-family ablation, a mass and inertia mismatch sweep, an unmodeled quadratic drag, and a band-limited non-periodic disturbance over five seeds.

The ablation bounds the role of each module. Observer-based compensation accounts for most of the error reduction, by roughly an order of magnitude under the periodic disturbance and by about seven times under the random one, and it remains effective for mismatches up to $\pm 50\%$ and for a drag reaching about 74% of the payload weight. The RBF term adds a consistent but modest 5.8–9.7%, and the ratio does not respond to the severity of the mismatch or of the unmodeled nonlinearity. The ordering of the four configurations is preserved in every disturbance class tested. The contribution claimed here is therefore the residual-consistent formulation, the conditional boundedness analysis of the payload layer, and a quantitative statement of what each module does and does not deliver, rather than a claim that the network learns the unmodeled dynamics.

Simulation Parameters

The physical, geometric, and numerical parameters are listed in Table 6; the single parameter set shared by all four configurations is listed in Table 7.

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Declaration of conflicting interests

The authors declare that there is no conflict of interest.

Table 6. Physical, geometric, and numerical parameters.

Parameter	Symbol	Value
Number of quadrotors	N	3
Integration step and duration	Δt, T_f	0.001 s, 20 s
Quadrotor mass	m_p	2.0 kg for each UAV
Quadrotor inertia	$\mathbf{J}_p$	$\mathrm{diag}(0.082, 0.0845, 0.1377)$ kg·m^2
Payload mass	m_L	2.0 kg
Payload inertia	$\mathbf{J}_L$	$\mathrm{diag}(0.113333, 0.173333, 0.273333)$ kg·m^2
Payload dimensions	$L \times W \times H$	$1.0 \times 0.8 \times 0.2$ m
Cable length	l_i	1.0 m for all three cables
Gravitational acceleration	g	9.81 m/s^2
Initial quadrotor positions	$\mathbf{p}_i(0)$	$[0.5, 4.2, 0]^T$, $[-0.5, 4.6, 0]^T$, $[-0.5, 3.8, 0]^T$
Initial payload position	$\mathbf{p}_L(0)$	$[0, 4.2, 0.1]^T$
Attachment points	$\boldsymbol{\rho}_i$	$[0.5, 0, 0.1]^T$, $[-0.5, 0.4, 0.1]^T$, $[-0.5, -0.4, 0.1]^T$

Table 7. The single parameter set shared by all configurations and all experiments. Only the compensation switches of Table 1 change between runs.

Group	Value
Payload backstepping gains	$\mathbf{K}_{p1} = \mathbf{K}_{p0} = \mathrm{diag}(3, 3, 3)$, $\mathbf{K}_{p2} = \mathrm{diag}(1.5, 1.5, 1.5)$; $\mathbf{K}_{\eta 1} = \mathrm{diag}(3, 3, 3)$, $\mathbf{K}_{\eta 2} = \mathrm{diag}(1.5, 1.5, 1.5)$
Allocation	$K_{\Omega q} = [43.7, 28.23, 17.54]$, $K_q = [50, 50.1, 50]$, $\alpha_f = 0.30$
UAV inner loop	backstepping thrust and moment controllers
RBF structure	6 channels $\times$ 5 nodes; centres $(0, 0)$, $(\pm 1, \pm 1)$; width $b = 0.8$; initial weights $\hat{\boldsymbol{\theta}}_i(0) = \mathbf{0}$
RBF scaling	translation $\mathrm{diag}(0.5\ \mathrm{m}, 0.5\ \mathrm{m/s})$; attitude $\mathrm{diag}(0.10\ \mathrm{rad}, 0.50\ \mathrm{rad/s})$
RBF adaptation	$\gamma_i = 20$, $\sigma = 0.05$, $\theta_{\max} = 5$, compensation scale 1.0
ESO	$\omega_0 = 50$ rad/s, i.e. $\boldsymbol{\beta} = [3\omega_0, 3\omega_0^2, \omega_0^3]$ in each block; compensation scale 1.0; 18 states; previous-interval applied wrench

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References

Bouabdallah S and Siegwart R (2005) Backstepping and sliding-mode techniques applied to an indoor micro quadrotor. In: *Proceedings of the IEEE International Conference on Robotics and Automation*. pp. 2247–2252. DOI:10.1109/ROBOT.2005.1570447.

Broomhead DS and Lowe D (1988) Multivariable functional interpolation and adaptive networks. *Complex Systems* 2: 321–355.

Estevez J, Garate G, Lopez-Guede JM and Larrea M (2024) Review of aerial transportation of suspended-cable payloads with quadrotors. *Drones* 8(2): 35. DOI:10.3390/drones8020035.

Fink J, Michael N, Kim S and Kumar V (2011) Planning and control for cooperative manipulation and transportation with aerial robots. *The International Journal of Robotics Research* 30(3): 324–334. DOI:10.1177/0278364910382803.

Gao Z (2003) Scaling and bandwidth-parameterization based controller tuning. In: *Proceedings of the American Control Conference*, volume 6. pp. 4989–4996. DOI:10.1109/ACC.2003.1242516.

Goodarzi FA, Lee D and Lee T (2015) Geometric control of a quadrotor UAV transporting a payload connected by flexible cable. *International Journal of Control, Automation and Systems* 13(6): 1486–1498. DOI:10.1007/s12555-014-0304-0.

Gu N, Wang D, Peng Z, Wang J and Han QL (2022) Disturbance observers and extended state observers for marine vehicles: A survey. *Control Engineering Practice* 123: 105158. DOI: 10.1016/j.conengprac.2022.105158.

Han J (2009) From PID to active disturbance rejection control. *IEEE Transactions on Industrial Electronics* 56(3): 900–906. DOI:10.1109/TIE.2008.2011621.

Lee T (2014) Geometric control of multiple quadrotor UAVs transporting a cable-suspended rigid body. In: *53rd IEEE Conference on Decision and Control*. pp. 6155–6160. DOI:10.1109/CDC.2014.7040353.

Lee T (2018) Geometric control of quadrotor UAVs transporting a cable-suspended rigid body. *IEEE Transactions on Control Systems Technology* 26(1): 255–264. DOI:10.1109/TCST.2017.2656060.

Lee T, Leok M and McClamroch NH (2013a) Geometric control of multiple quadrotor UAVs transporting a cable-suspended rigid body. *IEEE Transactions on Control Systems Technology* 21(4): 1341–1356.

Lee T, Leok M and McClamroch NH (2013b) Nonlinear robust tracking control of a quadrotor UAV on SE(3). *Asian Journal of Control* 15(3): 391–408. DOI:10.1002/asjc.567.

Lee T, Sreenath K and Kumar V (2013c) Geometric control of cooperating multiple quadrotor UAVs with a suspended payload. In: *52nd IEEE Conference on Decision and Control*. pp. 5510–5515. DOI:10.1109/CDC.2013.6760757.

Madani T and Benallegue A (2008) Adaptive control via backstepping technique and neural networks of a quadrotor helicopter. In: *IFAC Proceedings Volumes*, volume 41. pp. 6513–6518. DOI:10.3182/20080706-5-KR-1001.01098.

Madoński R and Herman P (2015) Survey on methods of increasing the efficiency of extended state observers. *ISA Transactions* 56: 18–27. DOI:10.1016/j.isatra.2014.11.008.

Park J and Sandberg IW (1991) Universal approximation using radial-basis-function networks. *Neural Computation* 3(2): 246–257. DOI:10.1162/neco.1991.3.2.246.

Sanner RM and Slotine JJE (1992) Gaussian networks for direct adaptive control. *IEEE Transactions on Neural Networks* 3(6): 837–863. DOI:10.1109/72.165588.

Shen S and Xu J (2021) Adaptive neural network-based active disturbance rejection flight control of an unmanned helicopter. *Aerospace Science and Technology* 119: 107062. DOI: 10.1016/j.ast.2021.107062.

Sreenath K, Michael N and Kumar V (2013) Trajectory generation and control of a quadrotor with a cable-suspended load–a differentially-flat hybrid system. In: *Proceedings of the IEEE International Conference on Robotics and Automation*. pp. 4888–4895. DOI:10.1109/ICRA.2013.6631275.

Thanh HLNN, Huynh TT, Vu MT, Mung NX, Tran HS, Kim SK, Vu TL and Nguyen NV (2022) Quadcopter UAVs extended states/disturbance observer-based nonlinear robust backstepping control under uncertainties and disturbances. *Sensors* 22(14): 5082. DOI:10.3390/s22145082.

Wang Z et al. (2022) Adaptive-based linear active disturbance rejection attitude control for a quadrotor UAV with external disturbances. *Transactions of the Institute of Measurement and Control* DOI:10.1177/01423312211031781.